



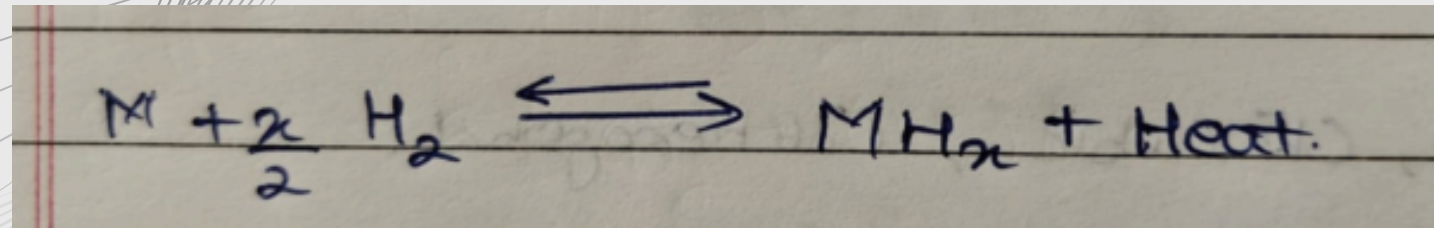
ME F366 – LABORATORY PROJECT

MADE UNDER THE GUIDANCE OF
PROF. SAURABH TIWARI

METAL HYDRIDE HYDROGEN COMPRESSORS (MHHC)

BY: KANISHQ GUPTA (F2022B2A41535G)

WORKING PRINCIPLE



1. Absorption (Low Temperature, Low Pressure):

- Hydrogen gas enters the hydride bed.
- Metal absorbs hydrogen forming a metal hydride.
- An exothermic reaction takes place, releasing heat and raising the temperature.
- Equilibrium pressure increases.

2. Desorption (High Temperature, High Pressure):

- Hydrogen is released at higher pressure from the hydride bed.
- Gas is delivered to storage or a downstream system.
- Hydride cools down and is ready for next cycle.

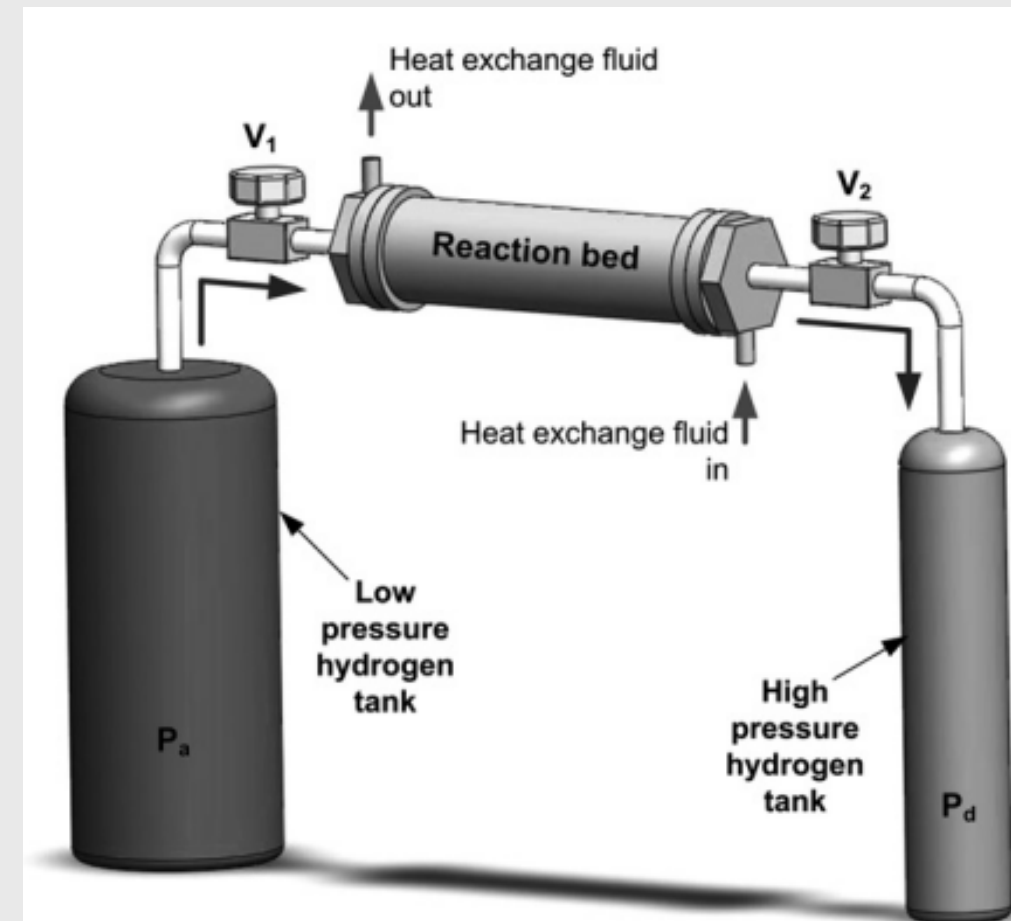


Fig. 1 – A schematic illustration of single stage MHTC.

Y. Wang et al., “Simulation study on the reaction process based single stage metal hydride thermal compressor,” *International Journal of Hydrogen Energy*, vol. 35, no. 1, pp. 321–328, Jan. 2010, doi: [10.1016/j.ijhydene.2009.10.085](https://doi.org/10.1016/j.ijhydene.2009.10.085).

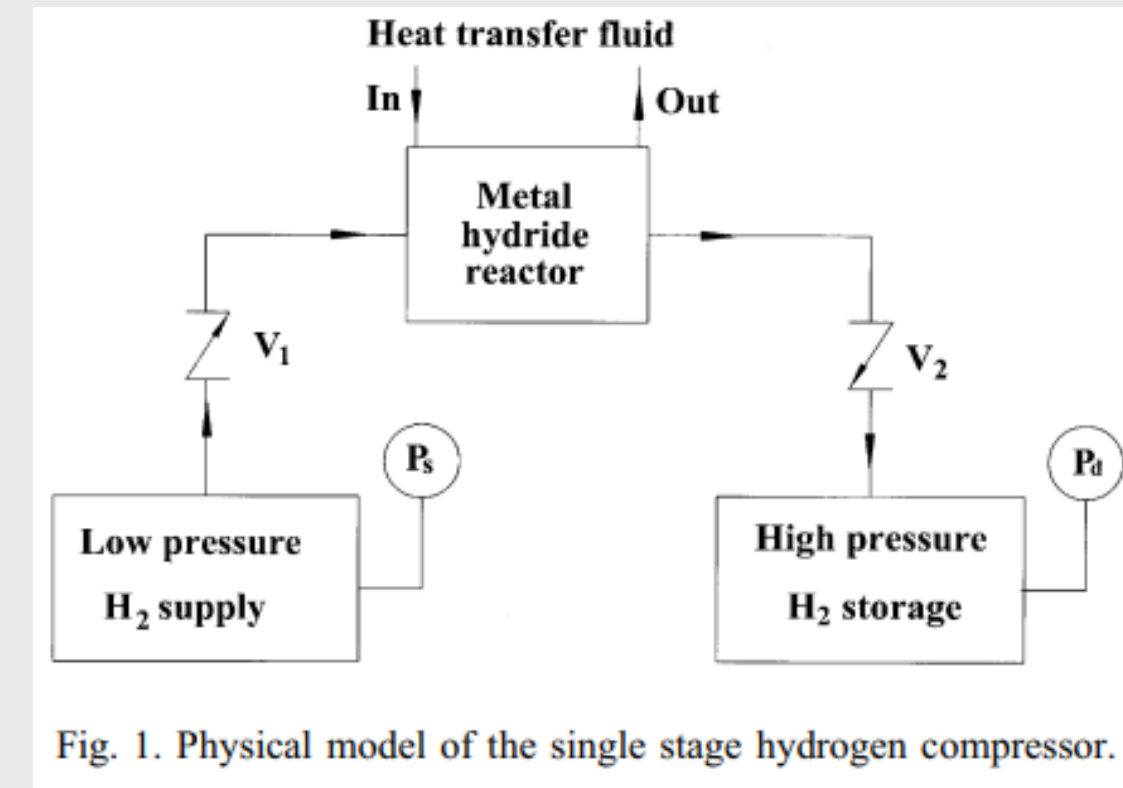


Fig. 1. Physical model of the single stage hydrogen compressor.

P. Muthukumar, M. P. Maiya, and S. S. Murthy, “Parametric studies on a metal hydride based single stage hydrogen compressor,” *International Journal of Hydrogen Energy*, vol. 27, no. 10, pp. 1083–1092, 2002, doi: [10.1016/S0360-3199\(02\)00005-8](https://doi.org/10.1016/S0360-3199(02)00005-8).

LITERATURE REVIEW

Simulation study on the reaction process-based single-stage metal hydride thermal compressor - Yuqi Wang

- The paper developed a thermodynamic model based MHHC for numerical simulations.
- It then studied the reaction fraction and temperature vs time graph for absorption and desorption conditions.
- It also studied the reaction fraction and temperature along the radial length graph for absorption and desorption conditions at different time stamps.

Alloy selection for dual-stage metal-hydride hydrogen compressor: Using a thermodynamic model to identify metal-hydride pairs - A Cousins

- The paper developed a mathematical model-based dual-stage MHHC for numerical simulations.
- The paper details the methodology and practices on deciding to choose which MHs to be used in case of more than one reactor involved.
- The paper also makes a mention of the various parameters that should be taken in account for while deciding the MHs used.

Computational study on metal hydride based three-stage hydrogen compressor - P. Muthukumar

- The paper developed a 3 stage MHHC model for numerical simulations using LaNi_5 , $\text{MmNi}_{4.6}\text{Al}_{0.4}$, $\text{Ti}_{0.99}\text{Zr}_{0.01}\text{V}_{0.43}\text{Fe}_{0.99}\text{Cr}_{0.05}\text{Mn}_{1.5}$.
- It provided graphs showing variation of average bed pressure and average hydrogen concentration with time.
- It showed that higher absorption temp worsens performance and lower temp ($\sim 20^\circ\text{C}$) improves hydrogen output.

Reactor design and numerical study on metal hydride based finned reactor configurations for hydrogen compression application - Abhishek Parida

- The paper developed a thermodynamic model based MHHC for numerical simulations.
- It provided a comparison between different fin configurations during absorption and desorption based on average bed temperature and concentration.
- It carried out an extensive performance study on the developed single-stage compressor.

OBJECTIVES

The primary objectives of this project were:

1. To study and understand the working principle of MHHC
2. To develop a mathematical model describing hydrogen absorption/desorption
3. To analyze the effect on temperature and pressure for metal hydride hydrogen compression applications.
4. To evaluate compression ratio and efficiency

MATHEMATICAL MODELLING

Equations related to mass conservation

Since porosity was constant, the mass conservation of hydrogen gas is given by Eq. (1) [41,43,54,55].

$$\varepsilon \frac{\partial \rho_g}{\partial t} + \nabla(\rho_g \cdot \vec{u}) = -S_{\text{mass}} \quad (1)$$

Similarly, the mass conservation of metal hydride is shown in Eq. (2) [54,55].

$$(1 - \varepsilon) \frac{\partial \rho_{\text{MH}}}{\partial t} = S_{\text{mass}} \quad (2)$$

Equations related to energy conservation

The energy conservation equation is given by Eq. (7) [41,43,54,55].

$$(\rho c_p)_{\text{eff}} \frac{\partial T}{\partial t} + (\rho c_p)_g (\vec{u} \cdot \nabla T) = \nabla(k_{\text{eff}} \cdot \nabla T) + S_{\text{heat}} \quad (7)$$

$$(\rho c_p)_{\text{eff}} = \varepsilon \cdot (\rho c_p)_g + (1 - \varepsilon)(\rho c_p)_{\text{MH}} \quad (8)$$

$$k_{\text{eff}} = \varepsilon \cdot k_g + (1 - \varepsilon)k_{\text{MH}} \quad (9)$$

The heat source term S_{heat} is given as Eq. (10) [41,43].

$$S_{\text{heat}} = S_{\text{mass}} \cdot \frac{\Delta H}{M_g} \quad (10)$$

$$\vec{u} = -\frac{\kappa}{\mu_g} \cdot \nabla P$$

$$S_{\text{mass}} = C_a \exp\left(-\frac{E_a}{R_u \cdot T}\right) \ln\left(\frac{P}{P_{\text{eq}}}\right) (\rho'_{\text{sat}} - \rho'_{\text{MH}}) \quad (3)$$

Here, ρ'_{sat} and ρ'_{MH} are effective densities of metal hydride bed at saturation and at any time instance as shown in Eqs. (4.1) and (4.2), respectively.

$$\rho'_{\text{sat}} = \rho_{\text{sat}}(1 - \varepsilon) \quad (4.1)$$

$$\rho'_{\text{MH}} = \rho_{\text{MH}}(1 - \varepsilon) \quad (4.2)$$

The saturated metal hydride density is calculated as $\rho_{\text{sat}} = \rho_i \cdot (1 + \text{wt}\%_{\text{sat}})$.

The equilibrium pressure for absorption, P_{eq} is determined by van't Hoff equation given as Eq. (5) [41,43,61].

$$P_{\text{eq}} = P_{\text{ref}} \exp\left[-\frac{\Delta H}{R_u T} + \frac{\Delta S}{R_u} + (\varphi + \varphi_0) \tan\{\pi(X - 0.5)\} + \frac{\beta}{2}\right] \quad (5)$$

$$\text{where, } X = \frac{\left(\frac{H}{M}\right)_t - \left(\frac{H}{M}\right)_i}{\left(\frac{H}{M}\right) - \left(\frac{H}{M}\right)} = \frac{\left(\frac{H}{M}\right)_t}{\left(\frac{H}{M}\right)} = \frac{\text{wt}\%_t}{\text{wt}\%_{\text{sat}}} \text{ [considering } (H/M)_i = 0].$$

$$\dot{m} = \begin{cases} C_a \exp\left(-\frac{E_a}{RT_s}\right) \ln\left(\frac{P_g}{P_{\text{eq}}}\right) (1 - X) : \text{absorption} \\ C_d \exp\left(-\frac{E_d}{RT_s}\right) \frac{P_g - P_{\text{eq}}}{P_{\text{eq}}} X : \text{desorption} \end{cases}$$

Initial conditions

At the beginning of the reaction, the density of metal hydride bed (ρ_{MH}) and the level of saturation (X) is assumed to be uniform throughout the bed.

$$\rho_{\text{MH}}(x, y, z, t=0) = \rho_i \quad (12)$$

$$X(x, y, z, t=0) = 0 \quad (13)$$

The cooling water domain is assumed to be at the supply temperature (T_s)

$$T(x, y, z, t=0) = T_s \quad (14)$$

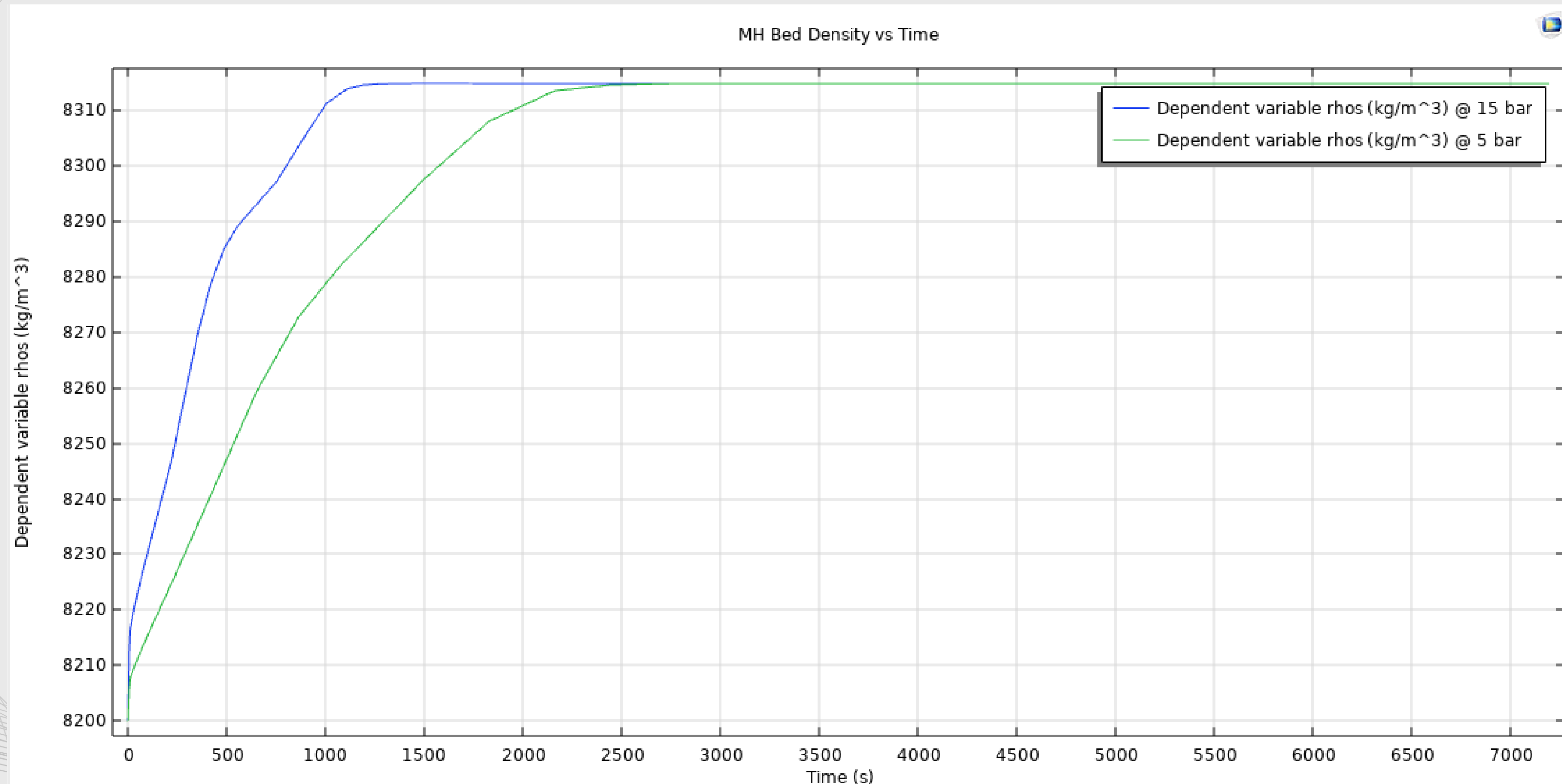
Everything except the water within the tank are assumed to be at the ambient temperature of 300 K.

$$T(x, y, z, t=0) = T_{\text{amb}} = 300 \text{ K} \quad (15)$$

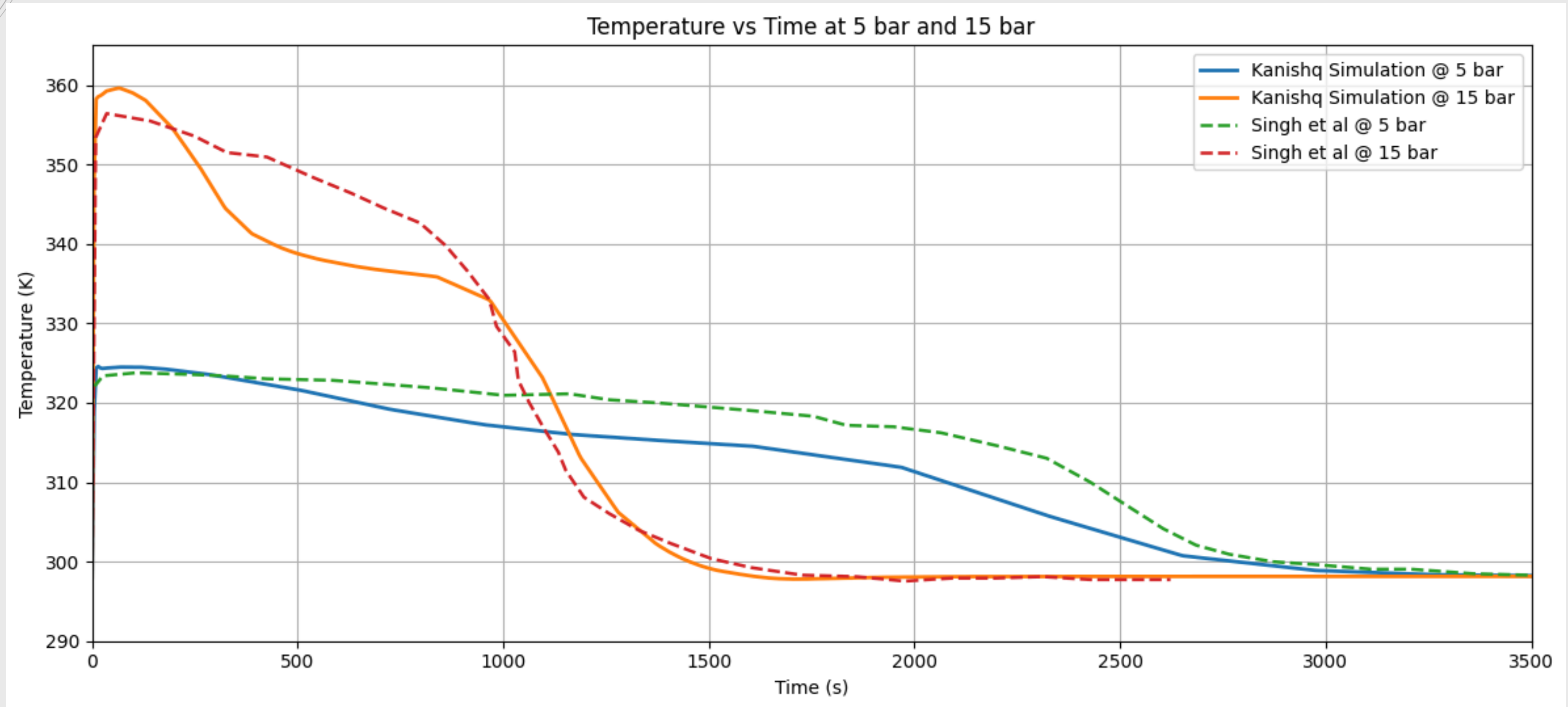
$$P(x, y, z, t=0) = P_i = P_s$$

S. Chandra, P. Sharma, P. Muthukumar, and S. S. V. Tatiparti, "Modeling and numerical simulation of a 5 kg LaNi₅-based hydrogen storage reactor with internal conical fins," *International Journal of Hydrogen Energy*, vol. 41, no. 30, pp. 13106–13123, 2016, doi: 10.1016/j.ijhydene.2016.05.140.

ABSORPTION SIMULATION RESULT



The model shows an average deviation of ~3.1%,
corresponding to an accuracy of approximately 97%.



CONCLUSION

The primary achievements of this project were:

1. A basic understanding of the working principle of MHHC was completed with a literature review.
2. A mathematical model describing hydrogen absorption in MHHC using LaNi_5 was developed, and a simulation was carried out on COMSOL to validate it.
3. The effects on temperature and pressure with time for metal hydride hydrogen compression were shown graphically at 5 bar & 15 bar pressures.

The possible future works of this project are:

1. Develop and simulate a mathematical model for desorption in MHHC using LaNi_5 .
2. To evaluate compression ratio and efficiency of the developed simulation.
3. Creation and development of a multi-staged MHHC model.
4. Validate and study the impact of other Metal Hydrides in the MHHC model.



THANK YOU